

Кладковская Мария 5130904/300002 Вар 47

Интерполяционный полином Эрмита

$$f(x) = \sin(\pi x); x \in [-\frac{1}{4}; \frac{1}{4}]; n=1$$

$$\frac{df}{dx} = \pi \cos(\pi x)$$

No	X	f(x)	$\frac{df}{dx} = m$
0	$-\frac{1}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$
1	$\frac{1}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2}$

$$h_1 = x_1 - x_0 = \frac{1}{4} - (-\frac{1}{4}) = \frac{1}{2}$$

$$t = \frac{x - x_0}{h_1} = \frac{x + \frac{1}{4}}{\frac{1}{2}} = \frac{4x + 1}{2}$$

$$H_1(t) = f_0(1 - 3t^2 + 2t^3) + f_1(3t^2 - 2t^3) + m_0 h_1(t - 2t^2 + t^3) + m_1 h_1(-t^2 + t^3)$$

$$H_1(t) = -\frac{\sqrt{2}}{2}(1 - 3t^2 + 2t^3) + \frac{\sqrt{2}}{2}(3t^2 - 2t^3) + \frac{\pi\sqrt{2}}{2} \frac{1}{2}(t - 2t^2 + t^3) + \frac{\pi\sqrt{2}}{2} \frac{1}{2}(-t^2 + t^3) =$$

$$= \sqrt{2}(6t^3 - 4t^2 - 1) + \frac{\pi\sqrt{2}(2t^3 - 3t^2 + t)}{2} = \frac{6t^3 - 4t^2 - 1}{\sqrt{2}} + \frac{\pi(2t^3 - 3t^2 + t)}{2\sqrt{2}}$$

$$H_1(x) = \frac{6(2x + \frac{1}{2})^3 - 4(2x + \frac{1}{2})^2 - 1}{\sqrt{2}} + \frac{\pi(2(2x + \frac{1}{2})^3 - 3(2x + \frac{1}{2})^2 + 2x + \frac{1}{2})}{2\sqrt{2}} = \sqrt{2}(3x - 16x^3) +$$

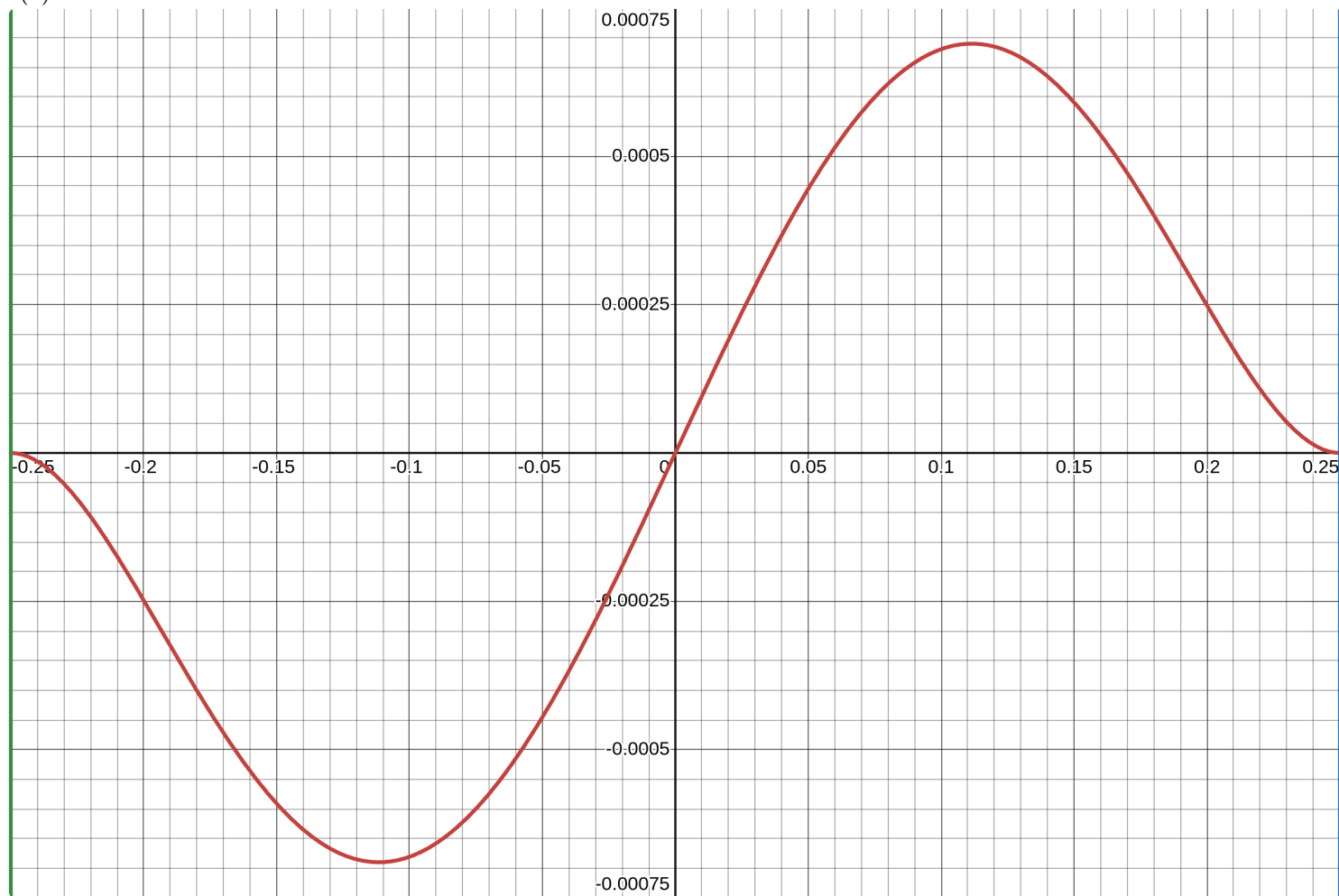
$$+ \frac{\pi(16x^3 - x)}{2\sqrt{2}} = x^3(-16\sqrt{2} + 4\pi\sqrt{2}) - x(3\sqrt{2} - \frac{\pi\sqrt{2}}{4})$$

График погрешности: $H_1(x) = T_1(x)$

$$r(x) = f(x) - T_1(x) = \sin(\pi x) - x^3(-16\sqrt{2} + 4\pi\sqrt{2}) - x(3\sqrt{2} - \frac{\pi\sqrt{2}}{4})$$

$$\frac{dr(x)}{dx} = \pi \cos(\pi x) - 3x^2(-16\sqrt{2} + 4\pi\sqrt{2}) - (3\sqrt{2} - \frac{\pi\sqrt{2}}{4})$$

$r(x)$:



$\frac{dr(x)}{dx}$:

